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Challenge Accepted: Turning AI's Massive Potential into Real Healthcare Value



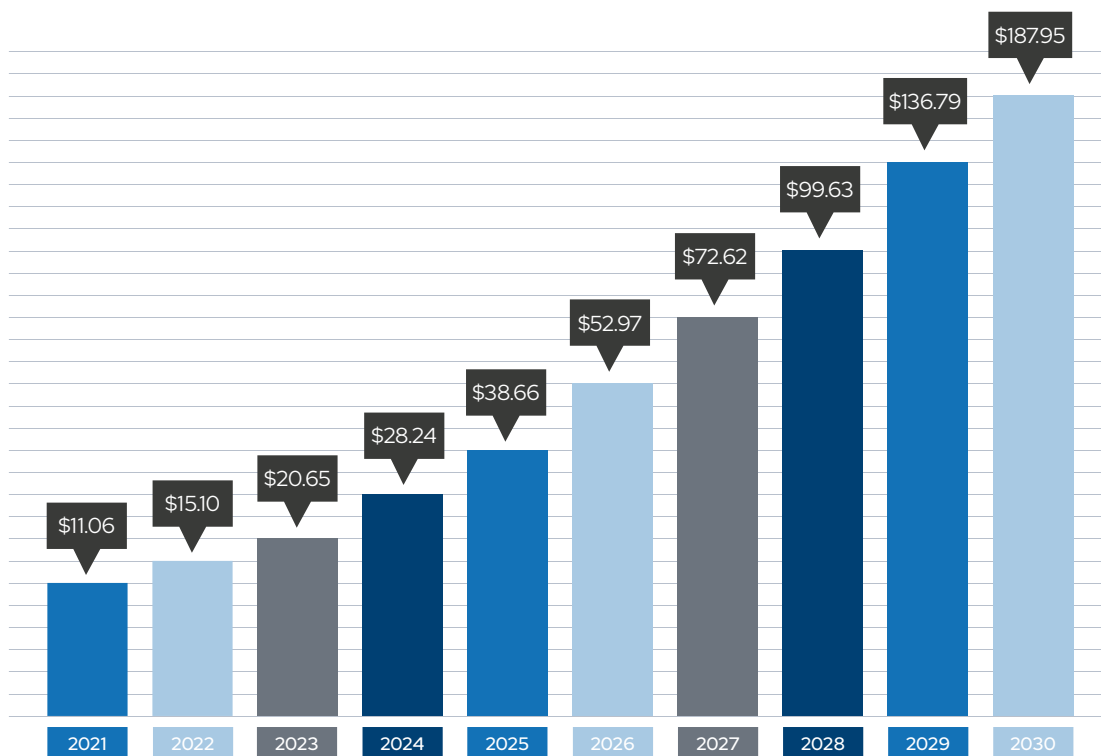


Today, medical research organizations serve as critical hubs for AI innovation. But what will it take to transform promising AI research into practical solutions that deliver reliable value for clinicians, healthcare organizations, and patients?

Artificial intelligence has become a top priority for anyone in healthcare who is responsible for leading or supporting technology innovation—especially within academic research organizations. Data scientists and medical researchers are actively working on advanced AI models to help clinicians identify an expanding range of serious diseases earlier, make healthcare operations more efficient, and bring clinical and administrative workflows together in important new ways. **As a result, investments in clinical AI technology are set to explode over the next few years—and drive a fundamental and far-reaching transformation in healthcare.**

Exploding investments in clinical AI

Artificial intelligence in healthcare market size, 2021 to 2030 (USD billion)



The size of the global AI in healthcare market is projected to grow from \$15.1 billion in 2022 to \$188 billion by 2030¹.



But we're clearly not there yet. AI adoption in healthcare lags behind other industries by significant margins². For many healthcare organizations, problematic gaps still exist between AI innovation and real-world impact. And in too many cases, promising AI models and research projects never complete (or even begin) their journey from research science to valuable application, because there is no systematic approach or strategy in place for piloting, validating, securing, scaling, and commercializing them.

Top concerns that are slowing AI adoption in healthcare²

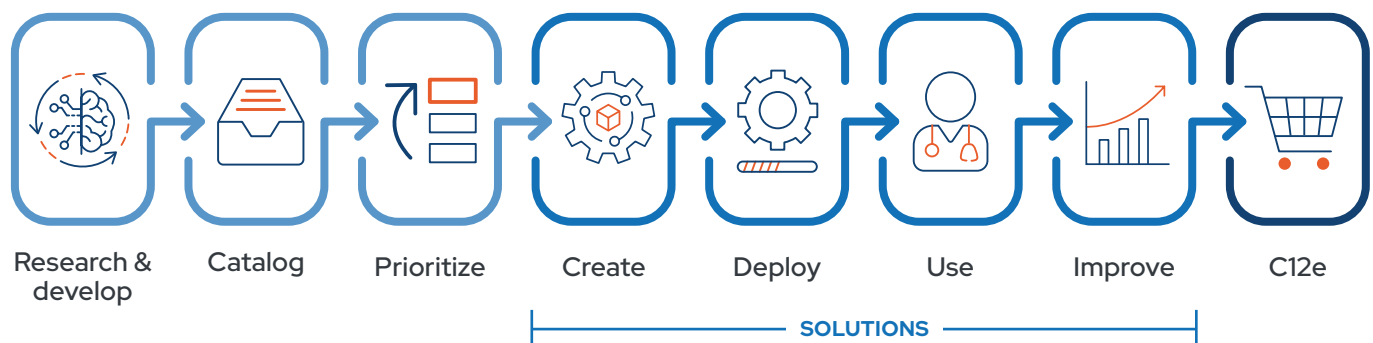
AI innovation in healthcare is accelerating, but the practical adoption of AI lags behind other industries.

Here are six reasons why:

1. Workflow integration challenges
2. Data sharing and privacy concerns
3. Transparency of algorithms
4. Data standardization
5. Interoperability across multiple platforms
6. Patient safety concerns

Before clinical AI can reach its full transformative potential, academic research centers and other healthcare organizations need to recognize these gaps, understand their root causes, and create cohesive, long-term strategies and roadmaps for reliably turning raw AI research and innovation into practical clinical solutions. Of course, that journey will look different for every organization—depending on size, type, mission, AI maturity levels, and many other factors. But there are a few well-tested principles and approaches that can guide and shape the development of any successful AI strategy. Here are a few things to consider.

The journey from AI innovation to long-term value





Catalog your IP: Know what you have and what you need

-  How much do you really know about all the different AI programs taking place inside your organization?
-  How many AI models have your research teams developed—including the ad-hoc models and algorithms that may be hidden away in obscure labs or inside larger projects?
-  How mature and effective are your AI assets?
-  Which programs and models have the greatest potential to impact patient outcomes?

These are relatively simple questions, but many research organizations genuinely struggle to answer them—mainly because their AI R&D programs are inherently decentralized and compartmentalized.

If this dynamic exists inside your organization, your AI strategy should initially focus on finding an efficient, centralized way to catalog and manage your entire AI portfolio. This makes it possible to holistically evaluate and prioritize your AI development, identify promising AI innovations earlier, find gaps in your portfolio that you may need to fill with external IP, and ultimately choose the best candidates for additional investment—including deployment and possible commercialization. Bringing all your AI assets together into a centralized, well-managed repository also opens the door to more advanced multi-modal, multi-model ways of thinking about AI that can make an even bigger impact on healthcare outcomes down the road.

Build a solutions-focused AI roadmap that aligns with your overall strategy

With a clear and complete picture of the AI-based IP at your disposal, you can start to build a smart, informed AI roadmap that aligns with your organization's overall mission and strategy. This involves identifying and prioritizing the models and projects in your AI catalog that best support your organizational goals. But it also includes developing a reliable, easily repeatable system for turning promising research into viable, real-world solutions. For many organizations, this will require building or strengthening new organizational muscles—or changing ingrained mindsets, attitudes, and incentive structures.

For example, credible academic research organizations already excel at funding, conducting, and publishing high-quality research. This process provides a powerful incubator for AI innovation—including the development of sophisticated AI models. But to become useful, these models must also be delivered successfully to the point of care. This requires multiple steps: integrating AI models naturally into trusted clinical workflows, deploying them successfully in real-world healthcare environments, continuously improving and refining them after deployment—just to name a few. To reliably make that crucial leap from promising innovation to proven clinical value, you need a cohesive, repeatable roadmap that starts with core model development and pilot programs and runs all the way through successful, long-term deployments—either by expanding your own in-house capabilities, working with a trusted partner, or both.



Make continuous improvement a core component of your AI roadmap

It's impossible to complete the journey from AI models to successful solutions without placing continuous learning and improvement at the center of your AI roadmap. At a basic level, continuous AI model improvement involves using new sources of data (including patient outcome data) to monitor, refine, and improve AI models after they're deployed. In other words, accessing and incorporating more outcome data from more sources provides a powerful tool for identifying gaps and limitations in datasets and algorithms, so researchers can make ongoing adjustments to improve the accuracy and effectiveness of their models over time.

Without these continuous model improvement processes, clinical AI models can't adapt to new or evolving sources of data, and they can't measure model-generated predictions against actual patient outcomes. As a result, models may become less accurate, and therefore less effective, over time—no matter how sound the algorithms or how much high-quality data was used to train them initially.

But continuous improvement doesn't end there. In solution-focused organizations, the concept of continuous AI improvement goes beyond the model itself to include other factors: How well are AI models integrated into established clinical workflows? Are AI insights being presented to clinicians or administrators in the most effective, useful way? How much of a measurable impact are AI predictions having on business operations or patient outcomes? An effective and complete continuous improvement strategy answers all these questions by treating AI models as one piece of a much broader overall solution—and actively working to improve every part of that solution.

Anatomy of a successful AI roadmap





Work to expand and evolve your best AI solutions

Turning a promising AI algorithm into a successful clinical or administrative solution is a notable accomplishment—but it doesn't necessarily mean you're getting the maximum value from that piece of intellectual property. In many cases, you can extend and enhance the value of successful commercial AI algorithms by adapting them to meet the requirements of different use cases. For example, with a solutions-focused AI roadmap in place, you could adapt an existing model that detects AFib in a broad, general patient population to focus on cancer patients who are at a higher risk of heart damage due to their chemotherapy treatments. Or, you could take a clinical model that only uses retrospective data and modify it to incorporate real-time patient data in the exam room to make more advanced predictions.

These types of creative, forward-thinking solutions often extend beyond the traditional realm of clinical model research and development. They require close collaboration among data scientists, software developers, healthcare providers, end users, and even patients. This broad, collaborative approach also sets the stage for innovative new multi-modal solutions, it allows you to tap into new sources of clinical and technological expertise, and it helps foster closer, more productive relationships with research partners. These more advanced benefits—which extend far beyond basic AI research and development—are only possible with a solutions-focused mindset backed by a holistic AI roadmap.

An expert's insight: What is an enterprise AI platform?

In my view, a platform is a combination of technology (data standards, APIs, security controls), policy (who can do what for what purpose with what privacy controls), and process (what workflow is supported by what people and what automation). In short, it is a way to use knowledge and technology to facilitate connections, and create value in the process.

— Dr. John Halamka, President, Mayo Clinic Platform

Embrace the advantages of a platform approach

Making the shift from AI model research to clinical solutions certainly isn't easy—especially when you consider the many additional steps and capabilities complete solutions require. AI research, software development, and IT deployment are all distinct disciplines—with different skillsets, priorities, and areas of focus. To successfully execute your AI strategy and roadmap, you'll need a multi-disciplinary team that can create a standard compute environment for your AI models, connect them to the right data resources, and deploy them in real healthcare environments quickly and consistently. You could try to assemble and



manage that team yourself. But for many organizations, it's more efficient to work with software engineering and deployment experts who already have the technical and industry expertise—along with a set of proven technology capabilities—to successfully deploy, scale, integrate, and continuously improve clinical AI solutions. This “platform approach” can play a key, central role in supporting, enabling, and accelerating the shift from a collection of ad-hoc AI models and research projects to viable, successful AI solutions.

As you evaluate your clinical AI platform options—and the role a clinical AI platform should play in your AI strategy—here are a few important capabilities your clinical AI platform should include:

Easy connections to diverse sources of healthcare data

Data forms the foundation of every successful AI implementation, and data quality correlates more closely with model performance than any other factor. But finding the right sources of clean, accessible, and well-governed data to support and improve your AI models can be a significant challenge—especially in healthcare organizations where data tends to be isolated in disparate, disconnected systems.

A capable clinical AI platform addresses this problem by making it easier to connect, integrate, and normalize data from a wide range of sources—from electronic health records to wearable devices. This includes harmonizing and standardizing diverse data formats and structures in ways that enable meaningful analysis and insights across complex healthcare ecosystems.

Robust security and data access policy controls

Few things matter more to healthcare organizations than data privacy and patient confidentiality. In today's highly regulated data environment, your clinical AI platform should provide robust security and privacy capabilities, including encryption, role-based access controls, audit trails, and anonymization techniques. As you leverage more sources of data to train, validate, and continuously improve clinical AI solutions, your clinical AI platform should enable and support your efforts to ensure data privacy, prevent unauthorized access, and meet the latest data security and patient privacy regulations.

This includes the ability to define and enforce granular policies on who can access specific data, for what purposes, and under what privacy controls.

Fast workflow integrations and adjustments

Healthcare workflows are dynamic, and they continuously evolve to adapt to changing needs. A clinical AI platform should address this challenge by supporting process automation and creating reusable components. This enables rapid adjustments to workflows, ensuring agility, efficiency, and scalability. Healthcare professionals can leverage predefined workflows, modify them as needed, and deploy them across the platform, resulting in streamlined processes and improved clinical outcomes.

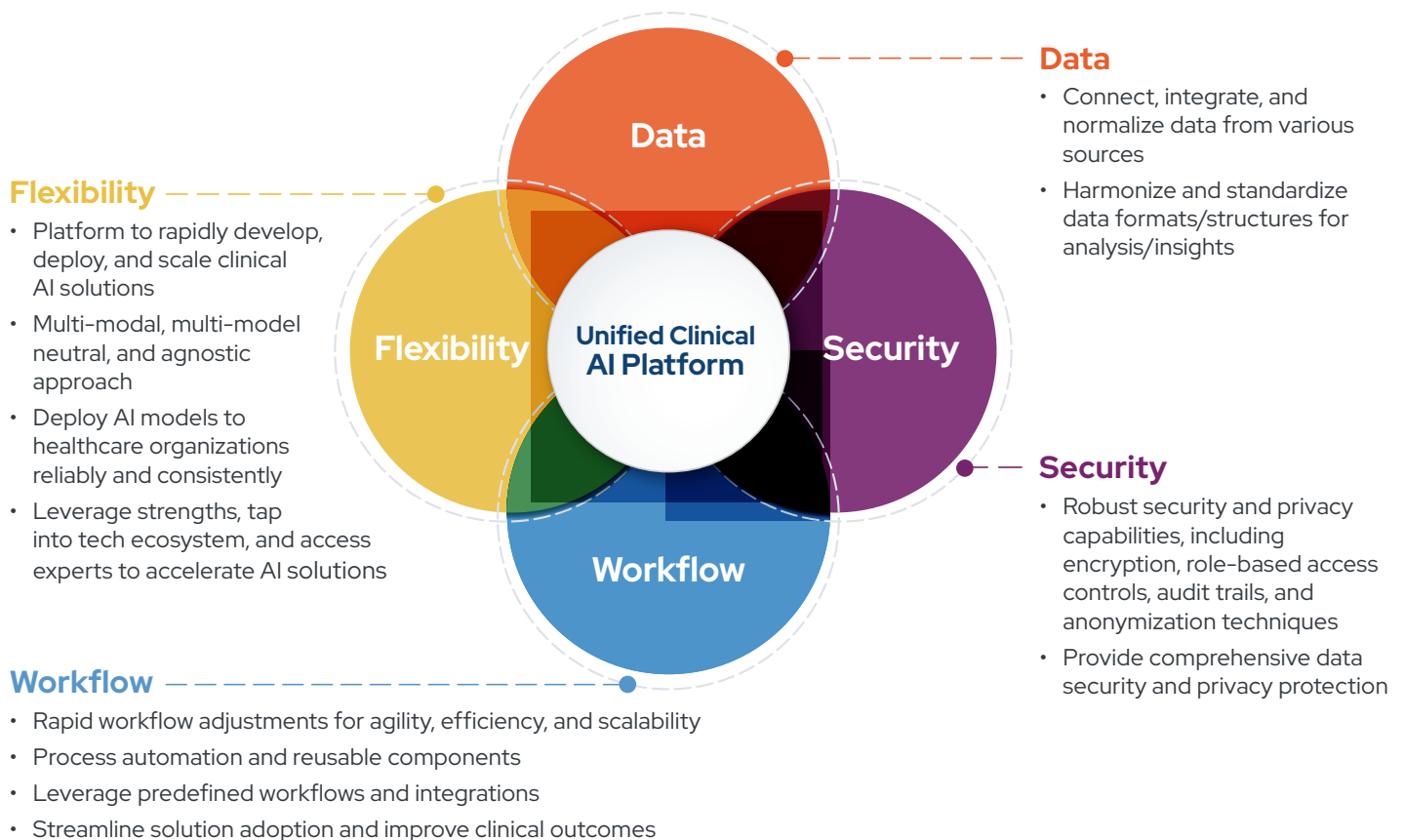


A flexible and comprehensive approach

Every clinical AI platform ultimately exists to turn meaningful AI research and innovation into viable, effective point-of-care solutions. This requires an approach that is both multi-modal and multi-model, which means it can integrate virtually any data with any algorithm—and any algorithm output with any workflow or application. It demands a neutral, open, and agnostic architecture that's capable of meeting the diverse needs of organizations across the innovation spectrum. And it should be designed to deploy AI models consistently and reliably to dozens or hundreds of healthcare organizations.

This type of open, comprehensive platform frees your researchers and data scientists to focus on what they do best—and makes it easy for them to tap into a powerful technology ecosystem and diverse team of experts to fill in the gaps, address key challenges, and provide the additional technologies and skillsets that will accelerate your journey from AI innovations to valuable clinical and operational solutions.

Turning AI innovations into successful solutions with an enterprise AI platform



An enterprise AI platform combines and unifies the technology, capabilities, and expertise you need to build, catalog, deploy, scale, and continuously improve complete clinical and administrative AI solutions.



Start accelerating your journey to AI value

With the current amount of interest and investment, no healthcare organization can afford to ignore the potential of clinical AI. With a smart, clear-eyed AI roadmap that recognizes the gaps and challenges you face—combined with a platform approach that makes the right mix of capabilities and expertise available to your organization—you are ready to turn your best AI innovations into solutions that deliver real value and lead to better healthcare outcomes.

Lucem Health: making better healthcare technology, together

Lucem Health creates productive, meaningful connections between the data scientists who develop predictive AI models and the clinicians who deliver front-line patient care. This makes it possible to bring transformational, AI-powered solutions from the lab to the front lines of healthcare—and provide clinicians with the context and understanding they need to trust, embrace, and value AI insights in their everyday work. Our approach is neutral by design. We don't make devices, algorithms, or applications. Instead, we unlock innovation by bridging the gaps between them.

www.lucemhealth.com

References

1. <https://www.precedenceresearch.com/artificial-intelligence-in-healthcare-market>
2. <https://www.nature.com/articles/s41591-018-0307-0>
3. <https://www.healthitanswers.net/what-is-a-platform/>